

Fuaad Maricar

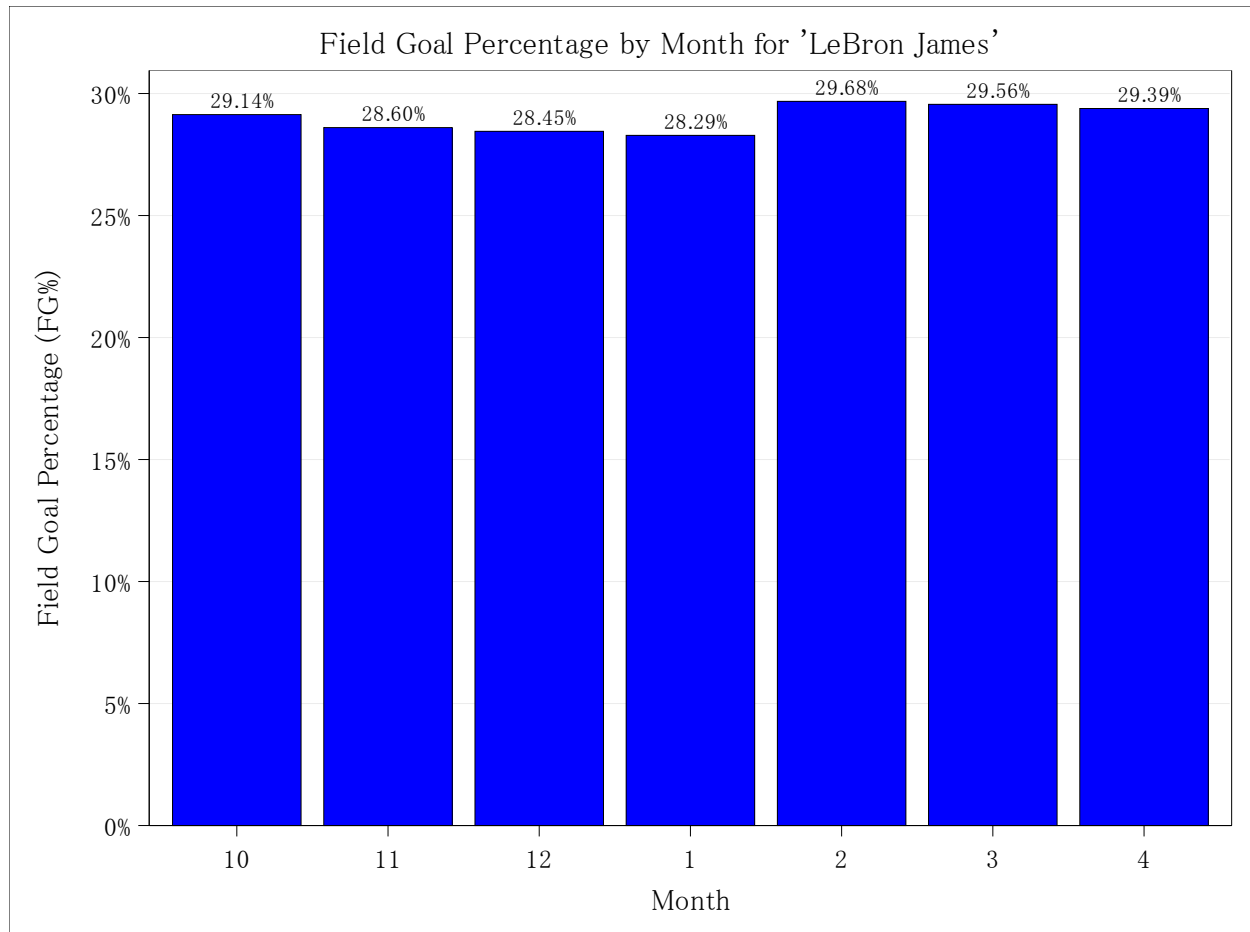
11 December 2024

STA 402 B Final Exam

Task: Write a SAS macro program that accepts a season and player name specified by the user and creates a graphical and summary analysis of the field goal percentage (FG%) of that player for that season. Include separate breakdowns of this statistic by month and opposing team.

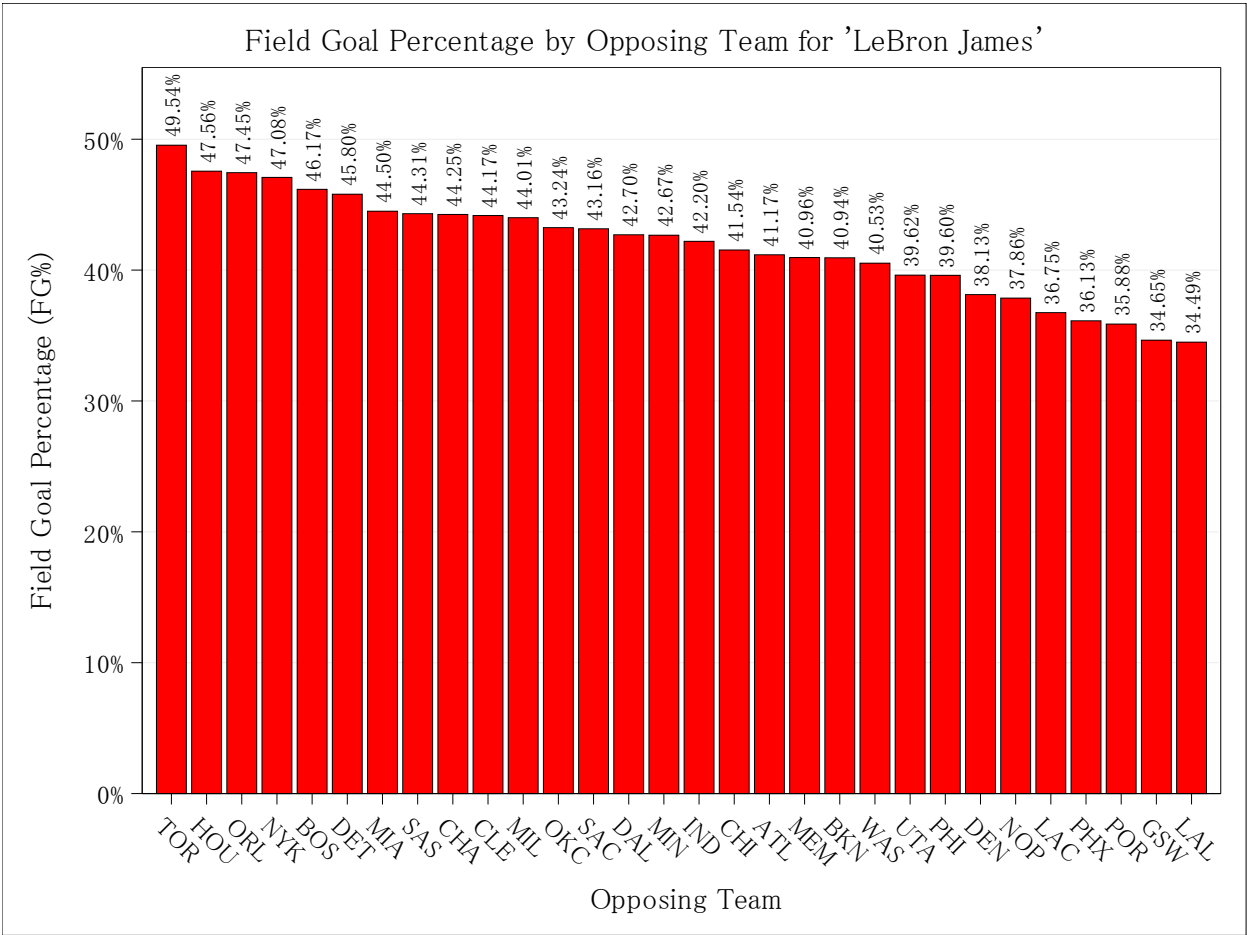
Output:

The first bar graph displays the field goal percentage by month for the specific player.



In this case the macro was run for the player "Lebron James" and season we are looking at is 2018-2019 and this graph displays Lebron's Field goal percentage by month in the specific season.

The second bar graph displays the output when a specific player’s Field goal percentage is compared to the opposing teams Field goal percentage. The opposing teams are all the teams the player played against throughout the season.



In this case “Lebron James” in the 2018-2019 season and his Field goal percentage compared to all the opposing teams he played against.

The following table just describes the results the bar graphs showed above.

Summary Statistics for FG% by Month for 'LeBron James'

month	Field Goal Percentage (FG%)
October	29.14%

month	Field Goal Percentage (FG%)
November	28.60%
December	28.45%
January	28.29%
February	29.68%
March	29.56%
April	29.39%

Code:

```
%let folder=C:\Users\fuaad\STA 402\STA402_final;
/*
Name: Fuaad Maricar
Date: November 4th, 2024
Final Project for STA 402
```

Purpose:

This SAS program defines a macro, `%FG\_analysis`, which analyzes basketball play-by-play data for the seasons 2015-2019. The macro generates a graphical summary of the Field Goal Percentage (FG%) for a specific player in a specified season. It also provides breakdowns of FG% by month and by opposing team. All output, including charts and tables, is saved to an RTF file.

Usage:

Make sure to change the file pathing to match with your machine up at the top before running the following macro.

To run the macro, specify the season (e.g., `2018-19`) and the player's name (e.g., `'LeBron James'`) as arguments:

Example: `%FG\_Analysis(season=2018-19, player='LeBron James');`

```
*/
```

```
/* File paths for datasets by season */
```

```
%let pbp_15 = &folder\2015-16_pbp.csv;
%let pbp_17 = &folder\2016-17_pbp.csv;
%let pbp_18 = &folder\2017-18_pbp.csv;
%let pbp_19 = &folder\2018-19_pbp.csv;
%let game_15 = &folder\gamedates_2015-16.csv;
%let game_17 = &folder\gamedates_2016-17.csv;
%let game_18 = &folder\gamedates_2017-18.csv;
%let game_19 = &folder\gamedates_2018-19.csv;
%let player_data = &folder\playerlist.csv;
%let play_by_play = &folder\README-basketball-play-by-play.txt;
```

```
%macro FG_analysis(season=, player=);
```

```
    /* Setting the proper file path based on the season */
```

```

        %if &season = 2018-19 %then %do; %let datefile = "&game_19"; %let
datafile = "&pbp_19"; %end;
        %else %if &season = 2017-18 %then %do; %let datefile="&game_18"; %let
datafile = "&pbp_18"; %end;
        %else %if &season = 2016-17 %then %do; %let datefile="game_17"; %let
datafile = "&pbp_17"; %end;
        %else %if &season = 2015-16 %then %do; %let datefile="game_15"; %let
datafile = "&pbp_15"; %end;
        %else %do;
            %put ERROR: We do not support that season.;
            %return;
        %end;

        /* Load play-by-play data for the selected season */
        data WORK.PBP_DATA ;
            infile &datefile delimiter = ',' MISOVER DSD lrecl=32767
firstobs=2 ;
            length v1-v17 $1 HOMEDESCRIPTION $77 PLAYER1_NAME $24
PLAYER1_TEAM_ABBREVIATION $3 PLAYER1_TEAM_CITY $13 PLAYER2_NAME $24
                PLAYER2_TEAM_ABBREVIATION $3 PLAYER2_TEAM_CITY $13
PLAYER3_NAME $24 PLAYER3_TEAM_ABBREVIATION $3 PLAYER3_TEAM_ABBREVIATION $3
PLAYER3_TEAM_CITY $13
                SCORE $9 SCOREMARGIN $3 VISITORDESCRIPTION $79
NEUTRALDESCRIPTION $1;
            input v1 EVENTMSGACTIONTYPE EVENTMSGTYPE v2 GAME_ID
HOMEDESCRIPTION NEUTRALDESCRIPTION v3 v4 v5 v6 v7 v8 PLAYER1_NAME
PLAYER1_TEAM_ABBREVIATION PLAYER1_TEAM_CITY v9 v10 v11 PLAYER2_NAME
PLAYER2_TEAM_ABBREVIATION PLAYER2_TEAM_CITY v12 v13 v14 PLAYER3_NAME
PLAYER3_TEAM_ABBREVIATION PLAYER3_TEAM_CITY v15 v16 SCORE SCOREMARGIN
VISITORDESCRIPTION v17;
            run;

        /* Load game dates for the selected season */
        data WORK.GAME_DATA;
            infile &datefile dsd firstobs=2;
            informat Date yymmdd.;
            input Game_ID Date;
            run;

        /* Filter data for the specified player and calculate FG attempts and
makes */
        data player_shot;
            set pbp_data;
            /* Define FG attempt and FG's made */
            if EVENTMSGTYPE = 1 then FGA = 1; else FGA = 0;
            if EVENTMSGTYPE = 1 and EVENTMSGACTIONTYPE in (1, 2, 3) then FGM
= 1; else FGM = 0;
            run;

        /* Calculate FG% by Month */
        proc sql;
            create table fg_month as
                select month(Date) as month,
                    sum(FGM) as FGM,
                    sum(FGA) as FGA,
                    calculated FGM / calculated FGA as FG_percent
format=percent8.2

```

```

        from player_shot join game_data on player_shot.Game_ID =
game_data.Game_ID
        group by month;
quit;

/* Calculate FG% by Opposing team. */
proc sql;
    create table fg_team as
    select PLAYER2_TEAM_ABBREVIATION as Opposing_Team,
           sum(FGM) as FGM,
           sum(FGA) as FGA,
           calculated FGM / calculated FGA as FG_percent
format=percent8.2
    from player_shot
    group by PLAYER2_TEAM_ABBREVIATION;
quit;

ods rtf bodytitle file = "&folder\Final_project.rtf";

/* FG% by Month - Bar Chart */
proc sgplot data=fg_month;
    vbar month / response=FG_percent datalabel
           fillattrs=(color=blue) stat=mean;
    yaxis label="Field Goal Percentage (FG%)" grid;
    xaxis label="Month" discreteorder=data values=(10 11 12 1 2 3 4);
    title "Field Goal Percentage by Month for &player";
run;

/* FG% by Opposing Team - Bar Chart */
proc sgplot data=fg_team;
    vbar Opposing_Team / response=FG_percent datalabel
           fillattrs=(color=red) stat=mean categoryorder=RESPDESC;
    yaxis label="Field Goal Percentage (FG%)" grid;
    xaxis label="Opposing Team" discreteorder=data;
    title "Field Goal Percentage by Opposing Team for &player";
run;

/* Ordering the months based on the season. */
proc format;
    value month_order
        10 = "October"
        11 = "November"
        12 = "December"
        1  = "January"
        2  = "February"
        3  = "March"
        4  = "April";
run;

/* Sorting months by the season */
data fg_month;
    set fg_month;
    if month < 10 then month_order = month + 20;
    else month_order = month;
run;

```

```
/* Assign the custom format and order by month */
proc sort data=fg_month;
    by month_order;
run;

/* Clean Summary Statistics Table for FG% by Month */
proc print data=fg_month label;
    id month;
    var FG_percent;
    format month month_order.;
    label FG_percent = "Field Goal Percentage (FG%)";
    title "Summary Statistics for FG% by Month for &player";
run;
ods rtf close;
%mend FG_analysis;
```